R108/B_density_ratio | TRAIN priors and actual latency

Complete forward including prior lookup and residual; no equal-gradient-scale or FLOPs claim.

Problem	Instances	Milliseconds / instance
TSP	128	0.249609
CVRP	128	0.298021
ATSP	128	0.069775
OVRP	128	0.069304
VRPB	128	0.069451
VRPL	128	0.069034
VRPTW	128	0.069242
OVRPTW	128	0.068997
OVRPB	128	0.069470
OVRPL	128	0.069160
VRPBL	128	0.069310
VRPBTW	128	0.069134
VRPLTW	128	0.069052
OVRPBL	128	0.069121
OVRPBTW	128	0.069331
OVRPLTW	128	0.069067
VRPBLTW	128	0.069090
OVRPBLTW	128	0.069488

Total trainable parameters: 1,842,953; product-noise binary density ratio on residual t

Optional R94B: not_requested

TRAIN-only prior SHA256: a763633f8fd7b7177c7a5a1574c6e31dcd6e3ba25cb6940e1372081f45ea23de

R108 uses the original scratch R84/R88B full18 selector with 1,842,953 trainables. Both arms share the same complete fresh tensors, fixed TRAIN-only (problem, exactN) prior buffers, forward computation and original maximin inference on s=logpi+t. Active gradient freedom is not identical: CE common-shift gauge cancels the shared last-bias data gradient, whereas the ratio anchors the utility origin. t is the original learned utility. A trains pure CE(s); B trains the canonical product-noise binary ratio: half positive softplus(-t_y) plus half pi-weighted softplus(t). Its normalization differs from CE; no equal gradient-scale claim. The ratio sigmoid is not a champion probability. Training classification losses and evaluation maximin CE are different diagnostics and are not compared as equal losses. Single forward, dropout0.1; comparison/risk/KL/RDrop weights and RDrop ramp are zero. 57,600 updates, logical batch1024, seed2, AdamW1e-4 to1e-6. Prior tables are small frozen aggregate TRAIN-fit classifier parameters, not per-instance features or query labels. Priors alone do not demonstrate instance learning: the saved VAL/TEST t=0 reference is a TRAIN winner-frequency conditional prior, not a cost-optimal SBS. Model gains versus that reference are reported separately from gains versus SBS. No new feature cache, donor, additional path module, teacher, context labels, new targets, calibration, quota or auxiliary loss. R89A AdamW, R89B Muon and R84A are saved historical references, not rerun or newly parameter/FLOP-matched controls; R94B is optional, never a prerequisite. Fixed36 rescues, losses, false picks and aligned net costs/regret are separate from choice frequency. Native/FP64 ind and legal pools; TRAIN-fitted SBS; ALL task-equal. TEST is development-exposed; one whole-model full18 VAL checkpoint for full TRAIN/VAL/TEST/LIB; no TEST/LIB reselection. Actual standalone GPU-resident-input timing includes complete model prior lookup and residual computation, excludes dataset I/O, CPU transfer and solvers, and is not end-to-end dataset latency. No routing-summary requirement. No online W&B equality/availability is asserted.

R108/B_density_ratio | Same selected model: full TRAIN

Diagnostic only; TRAIN was never used to select the checkpoint; full TRAIN-fitted SBS.

Problem	Top1 (%)	SBS Top1 (%)	mean_cost	SBS cost	Oracle cost	vs_SBS (%)	vs_Oracle (%)	actual regret (%)
ALL	71.4957	66.0332	11.389565	11.400702	11.345073	-0.1966	+0.3883	0.3979
TSP	46.4778	32.8617	8.004035	8.023959	7.970215	-0.2483	+0.4243	0.4215
CVRP	48.8506	18.1284	18.043487	18.151987	17.946816	-0.5977	+0.5387	0.7155
ATSP	77.6346	46.9210	1.639218	1.678679	1.633938	-2.3507	+0.3231	0.3222
OVRP	41.3815	34.2420	7.443178	7.453531	7.397676	-0.1389	+0.6151	0.6271
VRPB	80.6123	80.2765	9.361180	9.357264	9.319342	+0.0419	+0.4489	0.4616
VRPL	38.2704	31.5309	12.392349	12.400882	12.344503	-0.0688	+0.3876	0.3946
VRPTW	80.4346	80.2407	18.812422	18.813703	18.757954	-0.0068	+0.2904	0.3041
OVRPTW	83.8790	83.8025	10.118242	10.118640	10.093220	-0.0039	+0.2479	0.2438
OVRPB	88.4037	88.3580	6.363309	6.363230	6.346658	+0.0013	+0.2624	0.2704
OVRPL	41.9630	34.4716	7.443405	7.453219	7.398483	-0.1317	+0.6072	0.6180
VRPBL	79.9037	79.4593	9.404572	9.403115	9.363196	+0.0155	+0.4419	0.4466
VRPBTW	79.6568	79.4914	19.731607	19.733549	19.642712	-0.0098	+0.4526	0.4368
VRPLTW	80.4889	80.3827	18.768698	18.769959	18.714366	-0.0067	+0.2903	0.3046
OVRPBL	88.6037	88.5556	6.358846	6.358857	6.342513	-0.0002	+0.2575	0.2657
OVRPBTW	83.3642	83.2432	10.623487	10.624446	10.586225	-0.0090	+0.3520	0.3269
OVRPLTW	83.8235	83.7321	10.110489	10.111265	10.085768	-0.0077	+0.2451	0.2414
VRPBLTW	79.7630	79.5630	19.768730	19.770814	19.680364	-0.0105	+0.4490	0.4316
OVRPBLTW	83.4123	83.3370	10.624913	10.625544	10.587365	-0.0059	+0.3546	0.3304

ALL task-equal; ratios of mean costs differ from per-instance actual regret.

VAL ALL vs TRAIN frequency prior (not SBS), positive = better: Top1 gain +2.1173 pp; cost gain +0.0132281; regret gain +0.1085 pp.

TEST ALL vs TRAIN frequency prior (not SBS), positive = better: Top1 gain +2.0519 pp; cost gain +0.0128965; regret gain +0.1103 pp.

R108/B_density_ratio | Fixed target36 vs R89A

True rescues/losses, false picks and aligned net cost; fixed targets, no reselection.

Split	Winners	Rescued	Lost	New false	Net cost delta
TRAIN	83279	1330	93	1488	0.000452162
VAL	4625	70	6	89	0.000410146
TEST	4598	55	8	107	0.000719162

Split	Top1 delta pp	Corrected	Harmed	Cost delta	Regret delta pp
TRAIN	0.251235	32544	28881	0.000452162	0.003208
VAL	0.011111	1685	1676	0.000410146	0.000711
TEST	-0.014815	1678	1690	0.000719162	0.004908

Historical only, not retrained/matched; R89A AdamW, R89B Muon, R84A AdamW.

Fixed-target-only deltas do not replace the full18 population net cost/regret above.

R108/B_density_ratio | Fixed target36 vs R89B

True rescues/losses, false picks and aligned net cost; fixed targets, no reselection.

Split	Winners	Rescued	Lost	New false	Net cost delta
TRAIN	83279	1134	590	1382	0.001488489
VAL	4625	53	26	84	0.000948125
TEST	4598	47	29	100	0.001034196

Split	Top1 delta pp	Corrected	Harmed	Cost delta	Regret delta pp
TRAIN	-0.195405	28271	31120	0.001488489	0.014275
VAL	-0.186420	1563	1714	0.000948125	0.005715
TEST	-0.202469	1537	1701	0.001034196	0.009498

Historical only, not retrained/matched; R89A AdamW, R89B Muon, R84A AdamW.

Fixed-target-only deltas do not replace the full18 population net cost/regret above.

R108/B_density_ratio | Fixed target36 vs R84A

True rescues/losses, false picks and aligned net cost; fixed targets, no reselection.

Split	Winners	Rescued	Lost	New false	Net cost delta
TRAIN	83279	1361	90	1504	0.000404505
VAL	4625	73	5	89	0.000630932
TEST	4598	56	6	109	0.000661604

Split	Top1 delta pp	Corrected	Harmed	Cost delta	Regret delta pp
TRAIN	0.251852	32104	28432	0.000404505	0.002710
VAL	-0.023457	1647	1666	0.000630932	0.002897
TEST	-0.018519	1647	1662	0.000661604	0.004297

Historical only, not retrained/matched; R89A AdamW, R89B Muon, R84A AdamW.

Fixed-target-only deltas do not replace the full18 population net cost/regret above.